Q1. A scientist investigates the paints used in oil paintings. She takes tiny pieces of yellow, blue and green paint and tries to dissolve them in different solvents.
Her results are shown in the table.

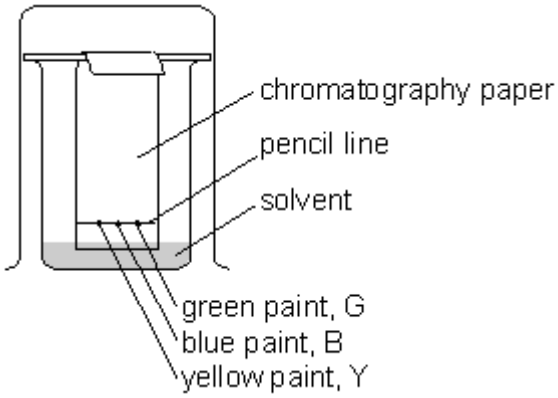
solvent	yellow paint	blue paint	green paint
water	yellow pieces are left	blue pieces are left	green pieces are left
ethanol	yellow pieces are left	clear blue liquid	clear blue liquid but yellow pieces are left
propanone	clear yellow liquid	clear blue liquid	clear green liquid

(a) Which solvent does **not** dissolve the blue paint?

.....

1 mark

She then uses chromatography to investigate the paints.



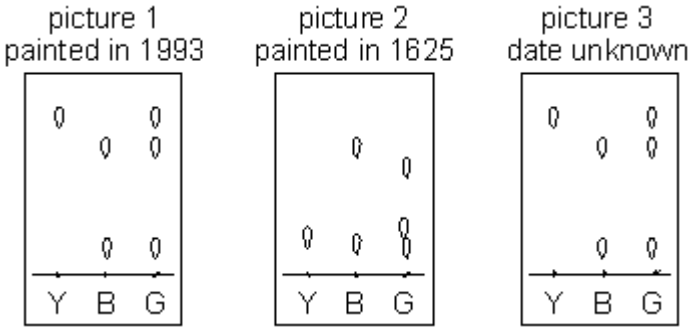
(b) Only **one** of the solvents in the table will make all three paints move up the chromatography paper. Which solvent is this?

.....

1 mark

(c) The scientist then investigates the paint used in three different oil paintings. She takes tiny pieces of yellow, blue and green paint from each picture and uses chromatography to compare them.

Her results are shown below:



Which of the paints in the 1993 picture contains only **one** substance?
Tick the correct box.

yellow, Y ☐ blue, B ☐ green, G ☐

1 mark

The scientist decides that picture 3 is probably recent and not from around 1625.

(d) Look at the chromatography results for the three pictures. Explain how the scientist was able to decide this.

.....

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.....

2 marks
Maximum 5 marks

Mark schemes

Q1.

(a) water

1 (L5)

(b) propanone

*do **not** accept 'propane' **or** 'propone'*

1 (L5)

(c) yellow ✓

if more than one box is ticked, award no mark

1 (L5)

(d) **one mark is for identifying that the same paints were used in picture 3 and picture 1, and the other is for identifying the differences between the paints used in picture 3 and picture 2**

any **one** from

- the three paint colours are all the same in pictures 1 and 3
- the pattern of colours is the same in picture 1 and 3
- the yellow and green paint are the same in pictures 1 and 3

accept 'picture 1 matches picture 3'

1 (L5)

any **one** from

- the yellow **or** green paint in picture 3 is different from that in picture 2
- only the blue is the same in pictures 2 and 3
- only the blue paint is the same in the three pictures

1 (L5)

[5]